

INSTRUCTION SHEET

Foam Core Bungalow

Build a one-storey scale model from a single sheet of foam core. Pick one of three designs (A, B, or C). Read this sheet completely before you cut anything.

MATERIALS & TOOLS

Materials (provided)	Tools (shared at workstation)
• 1 sheet foam core, 50 × 76 cm (5 mm thick) • Hot glue • Masking tape (for holding while glue sets)	• Ruler, 30 cm • Utility knife with fresh blade • T-square (for 90° corners) • <i>Bring your own:</i> pencil and eraser

SAFETY FIRST

Knife discipline. Cut away from your fingers, never towards them. Keep the hand holding the ruler with fingers together and back from the blade. Light pressure when guiding the blade along the ruler — too much and you'll shave the edge of the plastic. Change blades when they start to drag — a dull blade is more dangerous than a sharp one. If you ever lose the cap to your knife, stop and tell Mr. Feller immediately.

BUILD STEPS

01**Choose your design.**

Look at all three on the project website. Decide between A (Shed Roof), B (L-Shape), or C (Pitched Roof). Show your choice to Mr. Feller before continuing.

02**Plan your layout.**

On a scrap piece of paper, sketch how all your pieces will fit on the foam core sheet. Leave at least 5 mm between each piece for the blade. Don't waste material — you only get one sheet.

03**Measure and mark.**

Use a sharp pencil and your ruler. Mark every cut line on the foam core. Double-check every measurement against the cut list. Measure twice, cut once.

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- 04 Score the surface.**
With your knife and ruler, make a light first pass along each cut line. Don't try to cut all the way through on the first pass — you'll tear the foam and lose accuracy.
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- 05 Make the full cut.**
Run the blade through your scored line a second (and if needed, a third) time. Keep your ruler firmly in place. The piece should release cleanly. If it tears, your blade needs changing.
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- 06 Cut your door and windows.**
Mark these on the inside face of each wall before cutting. Cut along the longest sides first, then the short sides. Pop the cutout free.
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- 07 Dry-fit and mark.**
Stand all your walls up on the base **WITHOUT** glue. The walls sit ~1 cm in from each base edge, leaving a small lip all around. Use masking tape to hold them. Lightly pencil a line on the base around the inside edge of each wall — when you remove walls to apply glue, you can put them back exactly where they were. Adjust now, not after the glue dries.
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- 08 Glue the walls to the base.**
Apply a thin bead of hot glue along the bottom edge of each wall. Press firmly onto the base, lining up with your pencil marks. Use masking tape to hold each wall at 90° while the glue cools (about 1–2 minutes).
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- 09 Glue the walls to each other.**
Run a bead of glue along the vertical edge where two walls meet. Press together and hold. Check corners with your T-square — 90° matters.
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- 10 Attach the roof.**
Apply glue to the top edges of the walls. Lower the roof panel(s) carefully. For Design C, glue the two roof panels together at the ridge first, then lower the whole roof onto the walls.
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- 11 Final inspection.**
Look for gaps, weak corners, or wobbly walls. Reinforce anywhere that looks loose with extra glue from the inside. Trim any rough edges with your knife.
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- 12 Submit.**
Place your model on a sheet of paper with your name and design choice written on it. Hand in your reflection page. You're done.
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PRO TIPS

Glue sparingly. Too much glue oozes out the seams and looks messy. A thin bead is plenty.

Mind the table. You're cutting directly on the work table — keep cuts clean and don't press hard enough to score the surface underneath. Replace your blade as soon as it starts to drag.